Yuvraj Dutta

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EDUCATION

Master of Statistics (M.Stat.) — **2025–Present**
Indian Statistical Institute, Kolkata — First-Year Score: 92.0%
Dissertation Advisor: Prof. Sumit Mukherjee (Columbia University)

Bachelor of Statistics (B.Stat.) (Hons.) — **2022–2025**
Indian Statistical Institute, Kolkata — Aggregate Score: 90.3%
D. Basu Memorial Gold Medalist (Most Outstanding Student in B.Stat. Program)

RESEARCH INTERESTS

My research interests lie broadly in theoretical statistics, probability, and statistical machine learning, with particular interests in stochastic processes, statistical inference for diffusion and latent-variable models, graphical models, and interacting systems arising in statistical physics. More generally, I am interested in mathematically rigorous approaches to statistical inference and uncertainty quantification.

MANUSCRIPTS

Projection-Based Filtering and Smoothing of Diffusion Processes
Yannick Eich, Yuvraj Dutta, Heinz Koeppl
(Submitted to a ML/AI Conference)

Bivariate Inverse Gaussian Degradation Processes with Shared Random Effects and an Application to Fatigue Cracks
Yuvraj Dutta, Sandip Barui, Debanjan Mitra, Narayanaswamy Balakrishnan
`https://arxiv.org/abs/2606.04546`
(Submitted to Applied Stochastic Models in Business and Industry)

CURRENT RESEARCH

(Master's Dissertation) Asymptotic Fluctuations in Cubic Interaction Spin Models
Supervisor: Prof. Sumit Mukherjee — *Columbia University, New York*

Derived concentration inequalities for linear functionals in a cubic interaction spin model (which includes the edge-triangle ergm) using Chatterjee's exchangeable-pair framework and Glauber dynamics. Currently investigating CLT for the global magnetization under broad structural assumptions on the interaction tensor.

Conditional Limit Theorem for Brownian Motion with Drift under Restricted L^2-Norm
Supervisor: Prof. Frank Aurzada — *Technische Universität, Darmstadt*

Studying Brownian motion with drift conditioned to have an atypically small L^2-norm over a long time interval, generalizing recent results on rare-event conditioning for Brownian motion. A manuscript describing these results is currently in preparation.

SUMMER RESEARCH INTERNSHIPS

On-Site Research Intern — *May 2026 - Present*
Probability Group, Maths Department, TU Darmstadt, Germany
Supervisor: Prof. Frank Aurzada

On-Site Research Intern — *May 2025 - July 2025*
Self-Organizing Systems Lab, TU Darmstadt, Germany
Supervisor: Prof. Heinz Koeppl

Remote Research Intern *May 2024 - July 2024*
Interdisciplinary Statistical Research Unit, ISI Kolkata, India
Supervisors: Profs. S. Barui, D. Mitra, and N. Balakrishnan

INVITED TALKS AND PRESENTATIONS

Tracking the Hidden States : Bayesian Filtering & Smoothing
[Slides] / [Abstract] *D. Basu Memorial Gold Medal Award Seminar, ISI Kolkata, Oct 2025*

Conditional Limit Theorem for Brownian Motion with Drift under Restricted L^2-Norm
Summer Internship Research Presentation, TU Darmstadt, June 2026

Projection-Based Filtering and Smoothing of Diffusion Processes
Summer Internship Research Presentation, TU Darmstadt, July 2025

ACADEMIC ACCOLADES

Academic Achievements

- Awarded the **D. Basu Memorial Gold Medal**, presented annually to the **Most Outstanding Student in the B.Stat (Hons.) Programme** at the Indian Statistical Institute, Kolkata. *2025*
- Received the **Scholarship for Academic Excellence** in every semester of the B.Stat programme and M.Stat programme (till date), awarded for outstanding academic performance. *2022–26*

Mathematics & Competitive Examinations

- Ranked **8th nationally** and **24th overall** in the Pairs Category (with Nishant Lamboria) of the **Simon Marais Mathematics Competition** (East Division), an international undergraduate mathematics competition. *2024*
- Achieved **Regional Rank 11** in the **Madhava Mathematics Competition**, a nationwide proof-based mathematics competition for undergraduate students. *2023*
- Qualified for the **Indian National Mathematical Olympiad (INMO)**, after ranking among the top students regionally in the IOQM and qualifying for RMO each year . *2018–21*
- Secured **AIR 50** in the ISI B.Stat Entrance Examination, **AIR 1322** in JEE Main, and **AIR 1557** in JEE Advanced among 1 million+ candidates nationwide. *2022*
- Awarded the **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship**, a national scholarship programme for students demonstrating exceptional aptitude for scientific research. *2021*

SELECTED COURSEWORK PROJECTS

Statistical Depth and Robust Multivariate Inference [Report] *Nov 2025*

Model Diagnostics for Binary-Response GLMs: An NBA Case Study [Report] *Nov 2025*

Bootstrap-Based Model Selection in Binary GLMs [Report] *Mar 2026*

Studying Gameplay Factors via a Confounded Factorial Experiment [Report] *Apr 2026*

Resampling-Based Inference for Log-Linear Models [Report] *May 2026*

SKILLS

Programming	Python, R, C
Scientific Computing	NumPy, SciPy, PyTorch
Typesetting	LaTeX